

New explanations and inference for least angle regression

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- Tibshirani (1996) introduce Lasso.
- Efron et al. (2004) introduce Least Angle Regression (LAR).
- Dan and myself (2022–now) learn new things about LAR.

Least angle regression overview



- $\mathbf{y}, \mathbf{x}_1, \dots, \mathbf{x}_p \in \mathbb{R}^n$.
- Build predictor for $\mathbf{y}$ in steps:

$$\hat{\mathbf{y}}_0 \equiv \mathbf{0}$$

$$\vdots$$

$$\hat{\mathbf{y}}_k \equiv \hat{\mathbf{y}}_{k-1} + \text{update}_k$$

$$\vdots$$

$$\hat{\mathbf{y}}_p = \mathbf{P}_\mathbf{X} \mathbf{y}$$

- Each update_k is linear comb. of *active set* variables $\mathbf{x}_j \in \hat{\mathcal{A}}_k$.
- $\mathbf{P}_\mathbf{X}$ is orthogonal projection onto $\text{span}\{\mathbf{x}_1, \dots, \mathbf{x}_p\}$.

Least angle regression details



Lar($\mathbf{X}, \mathbf{y}$): Initialize $\hat{\mathbf{y}}_0 \equiv \mathbf{0}$ and update $\hat{\mathbf{y}}_{k-1}$ to $\hat{\mathbf{y}}_k$ by:

$$\hat{\mathbf{c}}_k \equiv \mathbf{X}^T(\mathbf{y} - \hat{\mathbf{y}}_{k-1})$$

Current “correlations”

$$\hat{C}_k \equiv \|\hat{\mathbf{c}}_k\|_\infty;$$

Max correlation

if $\hat{C}_k = 0$ set $\hat{\mathbf{y}}_k \equiv \hat{\mathbf{y}}_{k-1}$ and stop.

Stopping rule

$$\hat{\mathcal{A}}_k \equiv \{j : |(\hat{\mathbf{c}}_k)_j| = \hat{C}_k\}$$

Active set

$$\hat{\mathbf{X}}_k \equiv [\text{sign}((\hat{\mathbf{c}}_k)_j) \mathbf{x}_j, j \in \hat{\mathcal{A}}_k]$$

Signed active design

$$\hat{\mathbf{A}}_k^{-2} \equiv \mathbf{1}^T (\hat{\mathbf{X}}_k^T \hat{\mathbf{X}}_k)^{-1} \mathbf{1}$$

Angle

$$\hat{\mathbf{a}}_k \equiv \hat{\mathbf{A}}_k \hat{\mathbf{X}}_k (\hat{\mathbf{X}}_k^T \hat{\mathbf{X}}_k)^{-1} \mathbf{1}$$

Equi-angular direction

$$\hat{\gamma}_k \equiv \text{details omitted}$$

Distance

$$\hat{\mathbf{y}}_k \equiv \hat{\mathbf{y}}_{k-1} + \hat{\gamma}_k \hat{\mathbf{a}}_k$$

Updated predictor



- Obtain *step coefficients* $\hat{\mathbf{b}}_k$ (byproducts?) such that

$$\mathbf{X}\hat{\mathbf{b}}_k = \hat{\mathbf{y}}_k, \quad k = 1, 2, \dots$$

- These are sparse, i.e.

$$(\hat{\mathbf{b}}_k)_{\hat{\mathcal{A}}_k} = (\mathbf{X}_{\hat{\mathcal{A}}_k}^T \mathbf{X}_{\hat{\mathcal{A}}_k})^{-1} \mathbf{X}_{\hat{\mathcal{A}}_k}^T \hat{\mathbf{y}}_k$$

$$(\hat{\mathbf{b}}_k)_{\hat{\mathcal{A}}_k^c} = \mathbf{0}.$$

Diabetes data from Efron et al. (2004)



- $\mathbf{y}$ = continuous measure of disease progression
- $\mathbf{X}$ = subject covariates
- $n = 442$
- $p = 10$

Diabetes data

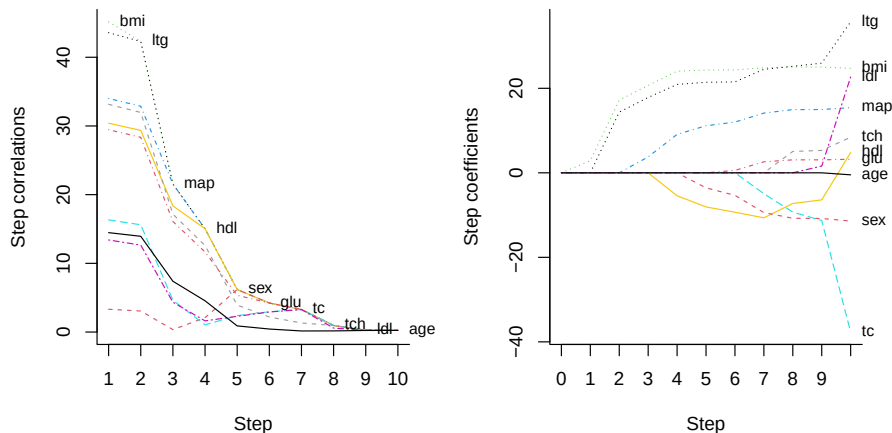


Figure: Step correlations $\hat{C}_k$, $\hat{c}_k$ and step coefficients $\hat{b}_k$.



- Lasso coefficients are

$$\hat{\mathbf{b}}(\lambda) \equiv \operatorname{argmin}_{\mathbf{b} \in \mathbb{R}^p} \|\mathbf{y} - \mathbf{X}\mathbf{b}\|^2 + \lambda \|\mathbf{b}\|_1, \quad \lambda > 0.$$

- $\hat{\mathbf{b}}(\lambda)$ piecewise linear with knots $\hat{\lambda}_1 > \hat{\lambda}_2 > \dots \geq 0$.
- Sometimes Lasso will coincide with LAR such that

$$\begin{aligned}\hat{\mathbf{b}}_k &= \hat{\mathbf{b}}(\hat{\lambda}_k) \\ \hat{C}_k &= \hat{\lambda}_k, \quad k = 1, \dots, p.\end{aligned}$$

- Lasso has possibly $> p$ steps (vars can leave active set).

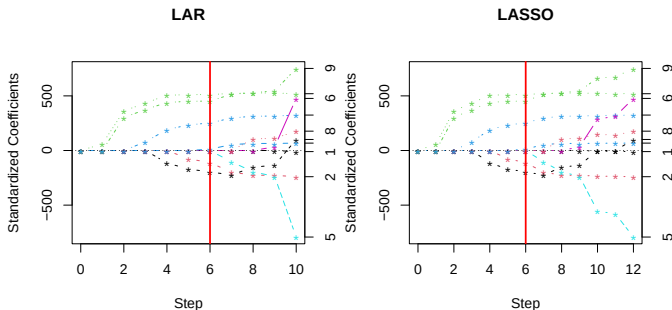


- A modification to LAR will yield the Lasso solution path.
- LAR has received most attention as a tool for analyzing Lasso.
- We focus on LAR itself.

Typical use of LAR or Lasso



- Obtain sparse estimator of β in $\mathbf{y} = \mathbf{X}\beta + \varepsilon$ by stopping midway:



- Inference hereafter is tedious.
- See post-selective work by Lee et al. (2016), Le Duy and Takeuchi (2021), and others.



Beyond sparsely estimating β in model $\mathbf{y} = \mathbf{X}\beta + \varepsilon$, we:

- Try to recover the complete path of LAR.
- Consider “population” path based on noiseless response $\mu = \mathbb{E}\mathbf{y}$.
- Estimate population counterparts $C_k, \mathbf{b}_k$ to $\hat{C}_k, \hat{\mathbf{b}}_k, k = 1, 2, \dots$
- Make inferences on these population counterparts.

Many sample path realizations

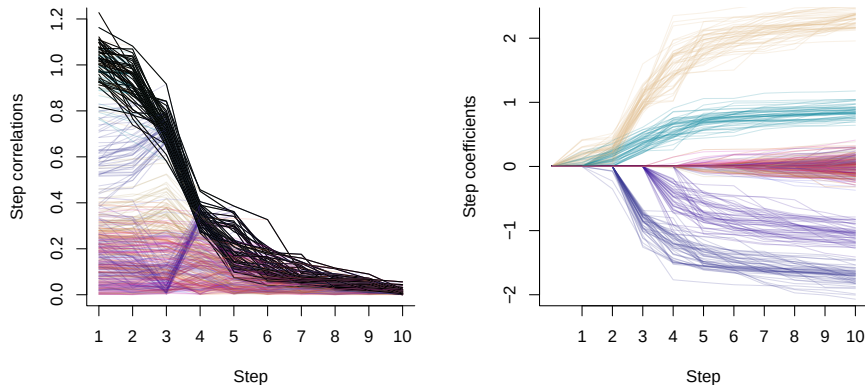


Figure: Simulated $\hat{C}_k$, $\hat{c}_k$ and $\hat{b}_k$ realizations at $n = 100$.

Will the sample path recover the pop. path?

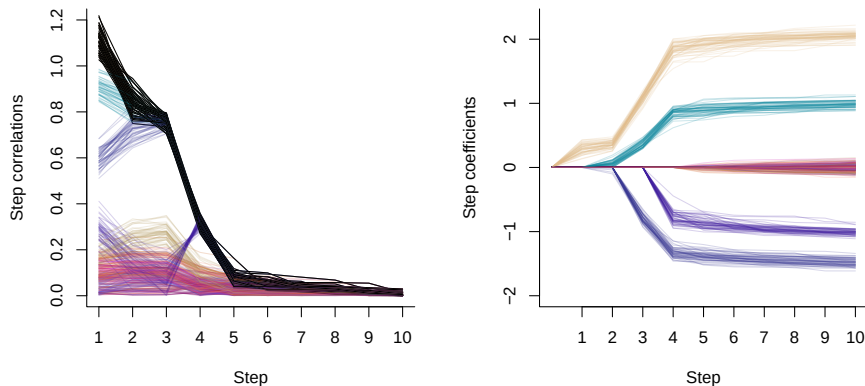


Figure: Simulated $\hat{C}_k$, $\hat{c}_k$ and $\hat{b}_k$ realizations at $n = 500$.



Lar($\mathbf{X}, \mu$), $\mu = \mathbb{E}\mathbf{y}$: Initialize $\mu_0 \equiv \mathbf{0}$ and update μ_{k-1} to μ_k by:

$$\mathbf{c}_k \equiv \mathbf{X}^T(\mu - \mu_{k-1})$$

Current “correlations”

$$C_k \equiv \|\mathbf{c}_k\|_\infty;$$

Max correlation

if $C_k = 0$ set $\mu_k \equiv \mu_{k-1}$ and stop.

Stopping rule

$$\mathcal{A}_k \equiv \{j : |(\mathbf{c}_k)_j| = C_k\}$$

Active set

$$\mathbf{X}_k \equiv [\text{sign}((\mathbf{c}_k)_j)\mathbf{x}_j, j \in \mathcal{A}_k]$$

Signed active design

$$A_k^{-2} \equiv \mathbf{1}^T(\mathbf{X}_k^T\mathbf{X}_k)^{-1}\mathbf{1}$$

Angle

$$\mathbf{a}_k \equiv A_k\mathbf{X}_k(\mathbf{X}_k^T\mathbf{X}_k)^{-1}\mathbf{1}$$

Equi-angular direction

$$\gamma_k \equiv \text{details omitted}$$

Distance

$$\mu_k \equiv \mu_{k-1} + \gamma_k\mathbf{a}_k$$

Updated predictor



- Population step coefficients are $\mathbf{b}_k$ such that

$$\mathbf{X}\mathbf{b}_k = \boldsymbol{\mu}_k, \quad k = 1, 2, \dots$$

- These are sparse, i.e.

$$\begin{aligned} (\mathbf{b}_k)_{\mathcal{A}_k} &= (\mathbf{X}_{\mathcal{A}_k}^T \mathbf{X}_{\mathcal{A}_k})^{-1} \mathbf{X}_{\mathcal{A}_k}^T \boldsymbol{\mu}_k \\ (\mathbf{b}_k)_{\mathcal{A}_k^c} &= \mathbf{0}. \end{aligned}$$



If $n \rightarrow \infty$ will we have:

- Step correlations and coefficients $\hat{C}_k, \hat{\mathbf{b}}_k$ close to $C_k, \mathbf{b}_k$?
- Active sets $\hat{\mathcal{A}}_1, \hat{\mathcal{A}}_2, \dots$ the same as $\mathcal{A}_1, \mathcal{A}_2, \dots$?
- Signed designs $\hat{\mathbf{X}}_k$ signed just as $\mathbf{X}_k, k = 1, 2, \dots$?



Definition (Prototypical LAR path)

Columns $\mathbf{x}_{j_1}, \dots, \mathbf{x}_{j_m}$ enter the active set one-by-one on steps $1, \dots, m$, resp. for some $m \leq p$ with $C_1 > \dots > C_m > C_{m+1} = 0$.

- Disallows “tied” entrances.
- Allows sparse representation of $\mathbf{P}_X \mu$.
- Original LAR seems designed with such paths in mind.
- Let $s_1, \dots, s_m$ denote the signings upon entrance of $x_{j_1}, \dots, x_{j_m}$.



Theorem 1 (Consistency of Lar($\mathbf{X}, \mathbf{y}$))

If for n large Lar($\mathbf{X}, \boldsymbol{\mu}$) is **prototypical** and satisfies a **design condition** and has **suff. separated entrances**, then under **regression assumps**:

- $P(\hat{\mathcal{A}}_k \setminus \hat{\mathcal{A}}_{k-1} = \{j_k\} \text{ and } \hat{s}_k = s_k) \rightarrow 1$
- $\hat{C}_k - C_k \xrightarrow{P} 0$
- $\max_{j \notin \mathcal{A}_k} |(\hat{\mathbf{c}}_k)_j - (\mathbf{c}_k)_j| \xrightarrow{P} 0$

for each $k = 1, \dots, m$ and

- $\hat{C}_k \xrightarrow{P} 0$

for each $k > m$ as $n \rightarrow \infty$ (p fixed).

Consistency under Theorem 1 conditions

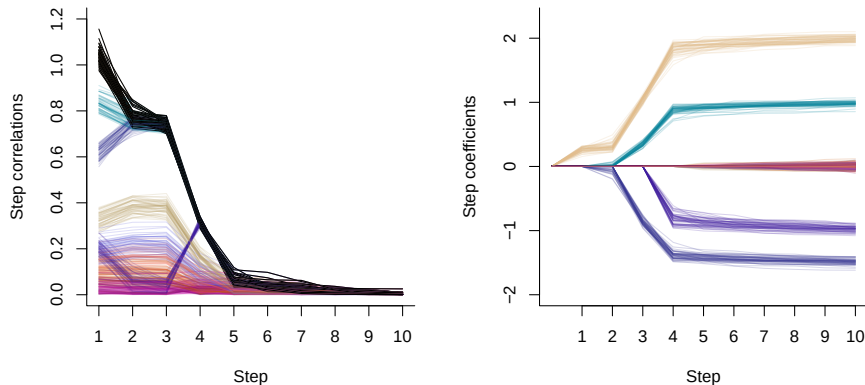


Figure: Simulated $\hat{C}_k$, $\hat{\mathbf{c}}_k$ and $\hat{\mathbf{b}}_k$ realizations at $n = 1000$.



- Variable to enter on step k is that which maximizes

$$C_{k,j}^2 = \frac{SS_{k,j}}{A_{j,k}^{-2} - A_{k-1}^{-2}},$$

where

$SS_{k,j}$ = Sequential SS for $\mathbf{x}_j$ given already-active variables

$A_{j,k}^{-2} - A_{k-1}^{-2}$ = Change in angle due to entrance of $\mathbf{x}_j$

- Strongest candidate:
 - (i) Reduces SS as much as possible.
 - (ii) Changes angle as little as possible.
- **Suff. separated entrances:** $\max C_{k,j}$ wins by a margin $\delta > 0$.

Pivotal quantity for step correlations



- For each sample step $k = 1, \dots, p$, define

$$T_{nk} \equiv \hat{s}_k \left(\frac{1}{\hat{A}_k^2} - \frac{1}{\hat{A}_{k-1}^2} \right)^{1/2} \sqrt{n} (\hat{C}_k - C_k),$$

setting $C_k \equiv 0$ for $k > m$.

- We derive the asymptotic joint distribution of $(T_{n1}, \dots, T_{np})$.
- Enables inference on $C_1, C_2, \dots$



Theorem 2 (Joint distribution of step correlation pivots)

Under Theorem 1 conditions

$$(T_{n1}, \dots, T_{np}) \xrightarrow{d} (T_1, \dots, T_p) \quad \text{as } n \rightarrow \infty,$$

where

- $(T_1, \dots, T_m)^T \sim \mathcal{N}(\mathbf{0}, \sigma^2 \mathbf{I}_m).$
- $(T_1, \dots, T_m) \perp\!\!\!\perp (T_{m+1}, \dots, T_p).$
- $\sum_{k=m+1}^p T_k^2 / \sigma^2 \sim \chi_{p-m}^2.$
- $(T_{m+1}, \dots, T_p) \sim \text{known (details omitted)}.$

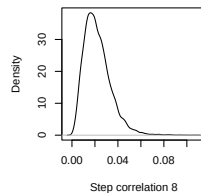
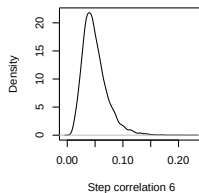
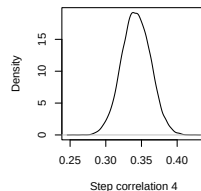
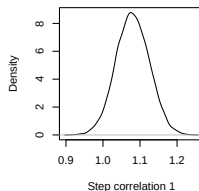
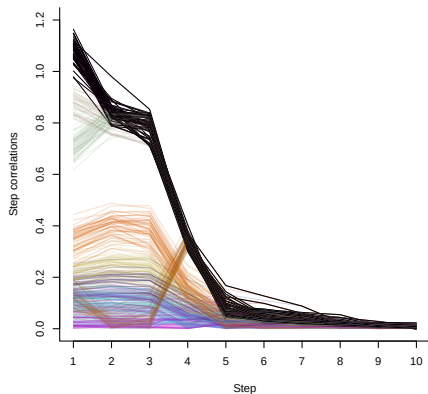


Figure: Simulated $\hat{C}_k$ realizations at $n = 500$.

Estimating the number of LAR steps



- Prototypical $\text{Lar}(\mathbf{X}, \mu)$ has $m \leq p$ steps.
- Use asymptotic result to construct an estimator for m .
- For $k = 1, \dots, p$ define statistics

$$\hat{W}_{nk} \equiv n \left(\frac{1}{\hat{A}_k^2} - \frac{1}{\hat{A}_{k-1}^2} \right) \frac{\hat{C}_k^2}{\hat{\sigma}^2}.$$

- Then $\hat{W}_{n1}, \dots, \hat{W}_{nm}$ are approx non-central χ_1^2 .
- While $\sum_{k=m+1}^p \hat{W}_{nk}$ is approx central χ_{p-m}^2 .

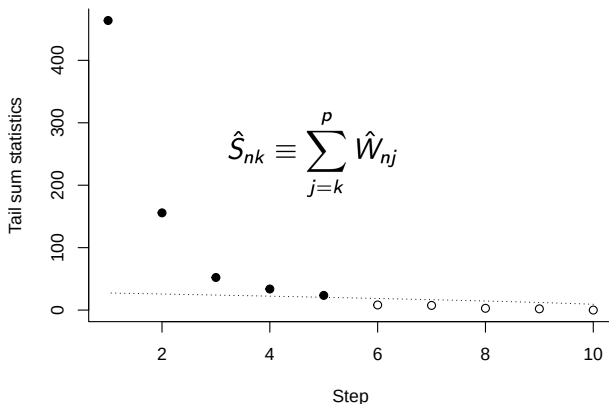


Figure: Finding $\hat{m}$ for diabetes data.

Corollary

Under Theorem 1 conditions $P(\hat{m} = m) \rightarrow 1$ as $n \rightarrow \infty$.

Diabetes data

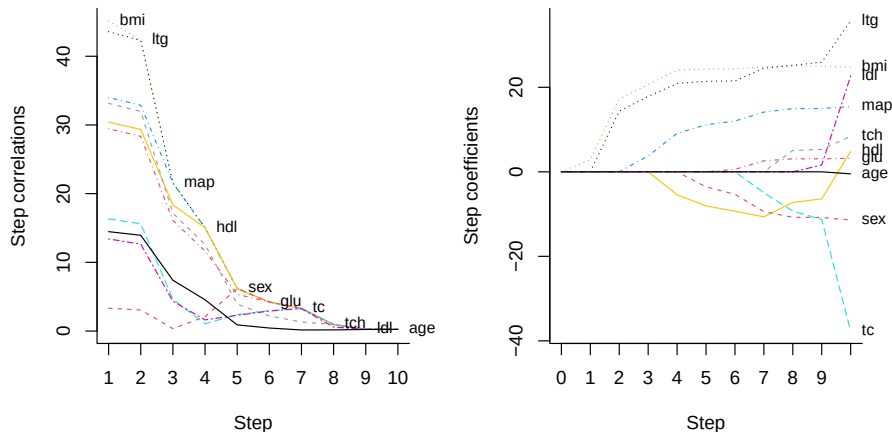


Figure: Now build confidence intervals for C_k , $\mathbf{b}_k$, $k = 1, \dots, \hat{m}$?

Diabetes data

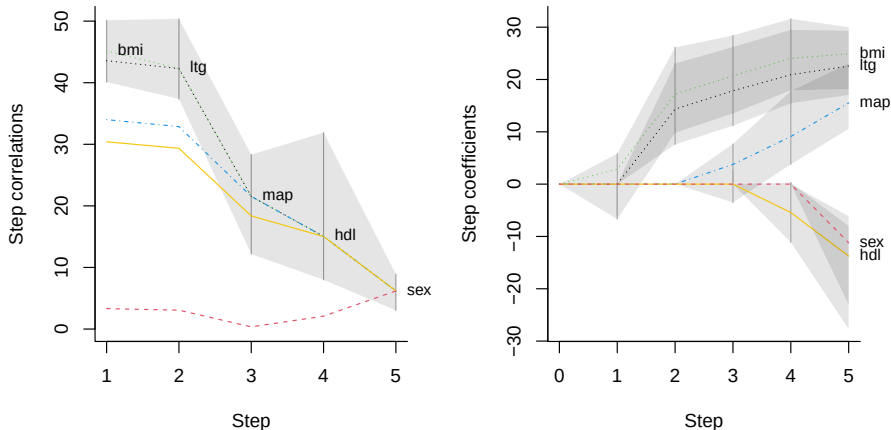


Figure: Bootstrap confidence intervals for C_k and $\mathbf{b}_k$, $k = 1, \dots, \hat{m}$.

Diabetes data



Variable	$\hat{S}_{nk}$	$\chi^2_{p-k+1, n-1}$	$\hat{C}_k$	2.5%	97.5%	$\hat{\mathbf{b}}_{\hat{m}}$	2.5%	97.5%
bmi	463.800	27.385	45.160	39.806	49.124	24.903	18.186	29.997
ltg	155.713	25.729	42.300	37.516	49.848	22.560	17.060	29.319
map	52.193	24.033	21.542	11.581	28.600	15.517	10.566	23.063
hdl	33.742	22.291	15.034	9.128	24.955	-13.752	-27.697	-8.015
sex	23.515	20.492	6.190	2.929	8.862	-11.215	-23.019	-6.119
glu	8.167	18.619	4.223	0.000	31.976			
tc	7.466	16.648	3.280	0.000	5.990			
tch	2.835	14.533	0.950	0.000	2.919			
ldl	1.994	12.183	0.261	0.000	0.607			
age	0.028	9.323	0.242	0.000	2.612			

Table: Summary of LAR inference on diabetes data

Performance in simulation of LAR inference



- Larger number of steps m is harder.
- Smaller margin $\delta > 0$ for **suff. separated entrances** is harder.
- Number of variables p not so important.

What about tied entrances?



- LAR can admit variables simultaneously.
- Never in sample path (i.e. with probability 0).
- Possibly in population path.
- If “ties” in population path, is sample path still consistent?

An unrecoverable population path



$\text{Lar}(\mathbf{X}, \mu)$: $\mathbf{x}_1$ enters on step 1; $\mathbf{x}_2, \mathbf{x}_3$ on step 2; $\mathbf{x}_4$ does not enter.

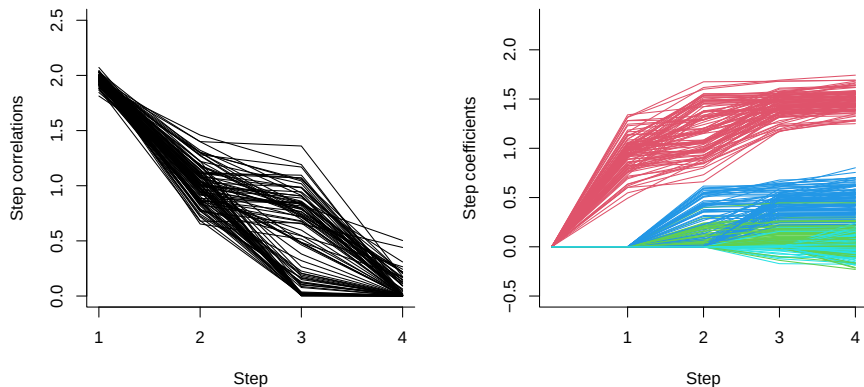


Figure: Many realizations of sample path $\text{Lar}(\mathbf{X}, \mathbf{y})$.

An unrecoverable population path



Population: $\mathbf{x}_1$ enters on step 1; $\mathbf{x}_2$, $\mathbf{x}_3$ on step 2; $\mathbf{x}_4$ does not enter.

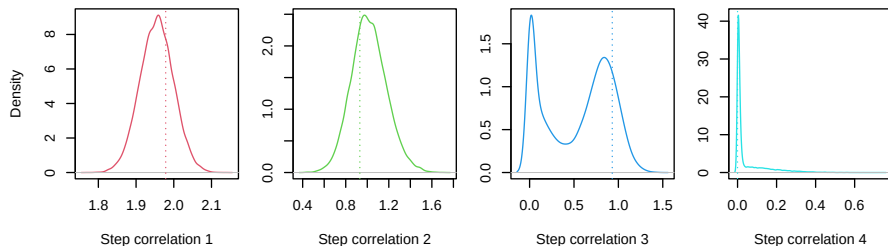


Figure: Distributions of $\hat{C}_1$, $\hat{C}_2$, $\hat{C}_3$, and $\hat{C}_4$ in a setting with mid-path ties.



- Case of tied entrances.
- Case when $p > n$.
- Translate results to the Lasso.



- Efron, B., Hastie, T., Johnstone, I., Tibshirani, R., et al. (2004). Least angle regression. *The Annals of statistics*, 32(2):407–499.
- Le Duy, V. N. and Takeuchi, I. (2021). Parametric programming approach for more powerful and general lasso selective inference. In *International conference on artificial intelligence and statistics*, pages 901–909. PMLR.
- Lee, J. D., Sun, D. L., Sun, Y., Taylor, J. E., et al. (2016). Exact post-selection inference, with application to the lasso. *The Annals of Statistics*, 44(3):907–927.
- Tibshirani, R. (1996). Regression shrinkage and selection via the lasso. *Journal of the Royal Statistical Society: Series B (Methodological)*, 58(1):267–288.



- ① **Design condition:** For some τ and ξ such that $0 < \xi \leq \tau < 1$, $|(\mathbf{c}_k)_j|/C_k \leq 1 - \tau$ for all $k \in \{2, \dots, m\}$ and $j \notin \mathcal{A}_k$ such that

$$\frac{|\mathbf{x}_j^T \mathbf{a}_k|}{A_k} < \frac{|(\mathbf{c}_k)_j|}{C_k} + \xi \quad \text{and} \quad \text{sign}(\mathbf{x}_j^T \mathbf{a}_k) = \text{sign}((\mathbf{c}_k)_j).$$

- ② **Sufficiently separated entrances:** For some $\delta > 0$, $C_1 - |(\mathbf{c}_1)_j| \geq \delta$ for all $j \neq j_1$ and $A_k(\gamma_{k,j} - \gamma_k) \geq \delta$ for all $j \notin \mathcal{A}_k \cup \{j_{k+1}\}$ and $k \in \{1, \dots, m-1\}$.

N.B. Distinct from the irrepresentable condition for Lasso.